Rohit Raj

Final Year Undergraduate
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Academic Qualifications

Year	Degree/Certificate	Institute
2021 - Present	B.S	Indian Institute of Technology, Kanpur
2021	BSEB(XII)	S.N.S. College,Gaya
2019	CBSE(X)	R.D Public School, Tekari

Scholastic Achievements

- Secured **All India Rank 2701** in **JEE Advanced 2021** among the 1.5 Lakh shortlisted candidates from across the country
- Secured **All India Rank 5842** in **JEE Main 2021** among approximately 9.3 Lakh participants from across the country

Key Projects

Hangman Word Guess *(May'24- Jun'24)*

Objective	• To create **Hangman word-guessing game** that predicts given secret word with a limited number of incorrect guesses.
Approach	• Developed a **9-gram** model on a 250,000-word dataset, and used statistical analysis to predict most probable letter. • Implemented a dynamic model using **weighted probabilities** from different n-grams to improve accuracy of the model. • Created a custom **Hangman game environment** which displays the status of the game after each guessed letter .
Impact	• Achieved approximately **60 % accuracy** for short words and **85 %** for long words, with up to 5 incorrect guesses.

SnapLink: A Fullstack Url Shortener | *Self Project* *Dec'23*

- Created a Full-Stack end to end URL management service, deployed backend on **AWS EC2** with a custom domain and SSL certification
- Developed a stunning frontend using **Nextjs** and **Tailwind CSS**, and authentication system using **Json Web Token** and **Nodemailer**
- Developed a robust Backend utilizing **Node.js** and **Express.js** leveraging **MongoDB** and **Mongoose** for efficient database management
- Used **Redux Toolkit** for state management and **RTK** query enabling efficient data fetching and streamlining the development

Realtime Facial Emotion Detection using Keras *(Jun'23-Jul'23)*

- Trained **convolutional neural network (CNN)** model with 28,821 training images taken from kaggle to recognize realtime facial emotions.
- Used **OpenCV** to automatically detect faces in images/webcam of a person and draw a bounding boxes(2 px) around the face.
- Tested the built model with seven different facial expressions using live webcam and got an accuracy of approximately **92%**.

Loan Eligibility Prediction *(May'23-Jun'23)*

- Built a predictive model to automate the process of predicting loan status and eligibility of an applicant for the loan with **12+ features**.
- Implemented the binary classifier models such as **Logistic Regression, Decision Tree** and **XGBoost** on dataset of above 900 clients.
- Trained all the models on stratified 5-fold cross-validated dataset and used **ROC curve** and confusion matrix for the model selection.
- Enumerated precision results and achieved the highest **F1-score of 0.79** and **AUC score of 0.74** with Random Forest Classifier model.

Metro App | *Course Project* *(May'23-July'23)*

- Made a **CPP** program that takes source and destination information as input and displays shortest distance and corresponding metro route.
- Used different algorithms like **Dijikstra, breadth-first search** and **depth-first search** to find shortest path between given metro stations.

Decentralised Certification | *Kerala Blockchain Academy* *(Jun'24- July'24)*

- Developed and deployed smart contracts for certificate issuance and governance on **Ethereum** blockchain using **Solidity** and **Hardhat**.
- Implemented **decentralized governance** with MyGovernor.sol and TimeLock.sol, enabling secure decision-making and platform updates.
- Conducted thorough testing and validation with Hardhat and JavaScript, managed dependencies using **pnpm** for efficient development.

Reinforcement Learning Quest | *Course Project* | *Prof. Subrahmanya Peruru* *(Jan'24-Apr'24)*

- Developed and implemented **Policy Iteration** and **Value Iteration algorithms** to optimize navigation in a stochastic maze environment.
- Implemented and compared **SARSA** and **Q-learning algorithms** to analyze their performance in a cliff walking grid world environment.
- Implemented **policy gradient algorithm** for balancing a cart pole,variants with and without a baseline, and evaluated their performance.

Technical Skills

- **Programming Languages and Utilities:** C, C++, HTML, LaTeX, MATLAB, Python, Java, SQL, PostgreSQL,Power BI, Tableau
- **Software and Libraries:** Github, Canva, AutoCAD, Numpy Pandas, Matplotlib, Scikit learn,Pytorch, OpenCv, TensorFlow

Relevant Courses

Data Structure and Algorithms	Probability and Statistics	Fundamentals of Computing
Introduction to RL	Non-Linear Regression	Human Computer Interaction

- **Attended Amazon ML Summer School 2024:** Completed an intensive program covering key Machine Learning topics such as **Supervised Learning, Deep Neural Networks, Dimensionality Reduction, Unsupervised Learning, Generative AI and LLMs, Sequential Learning, Causal Inference, and Reinforcement Learning**.

Positions of Responsibility

Organiser, Marketing, Udghosh'23 - *19th Edition: Annual Sports Festival of IIT Kanpur* *(2023-24)*

Leadership	• Spearheading a 3-tier team of 20+ senior executives and 40+ junior executives in the festival. • Led the marketing team of Udghosh for the Campus Ambassador program and successfully secured sponsors.
Initiatives	• Identified potential sponsors and successfully converted 10+ sponsors for the CA(Campus Ambassador) program. • Developing, compelling sponsorship proposals, showcasing the unique benefits for potential sponsors. • Securing valuable sponsorships from esteemed organizations, which contributes to the fest's financial success. • Successfully created a comprehensive database of companies to approach for sponsorship opportunities